

CRI Bioinformatics Bootcamp 2026

Day 6: Introduction to Spatial Transcriptomics

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Introduction to spatial transcriptomics: Visium, Xenium, CosMx



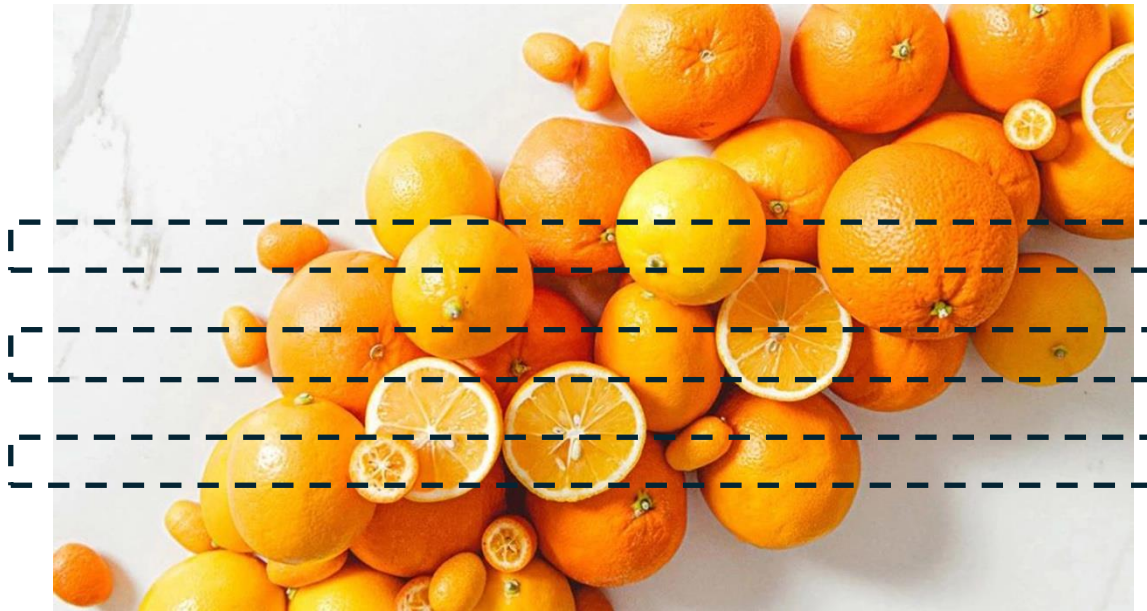
Bulk RNA



ScRNA-seq



*Spatial
Transcriptomics*

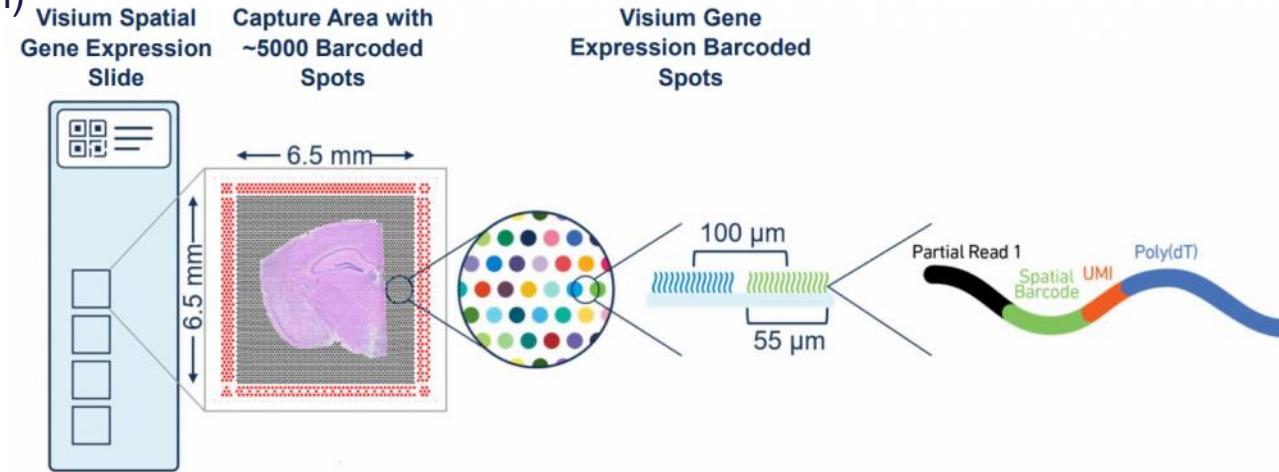


Serial sections will have different cells/parts of cells based on the snapshot in time of the bulk tissue

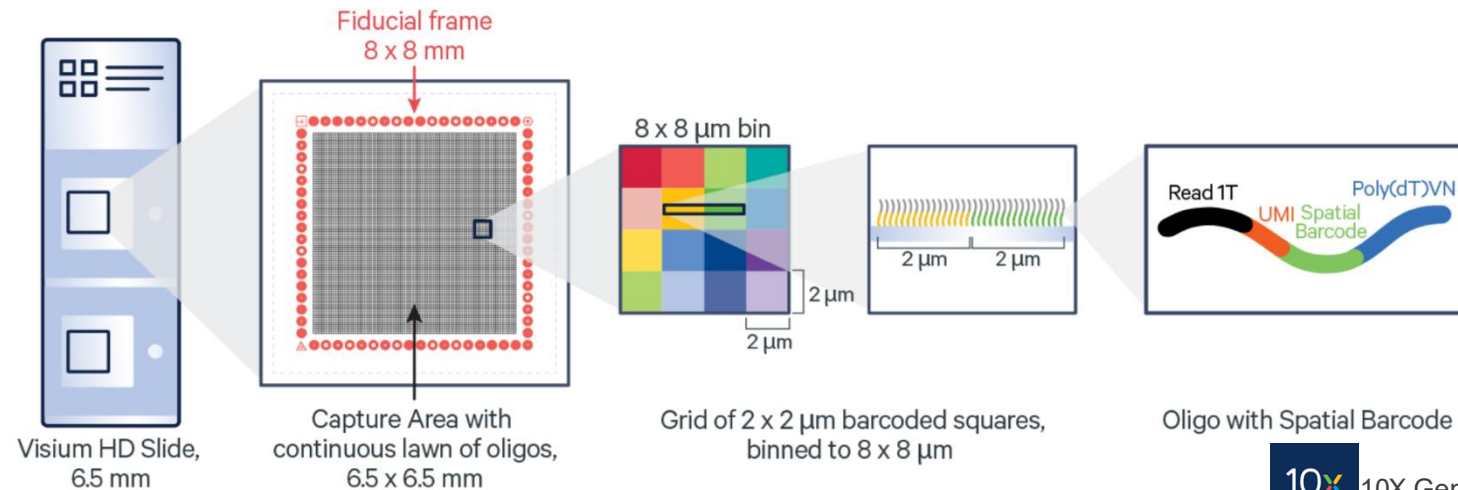
Visium HD

- Material (FFPE 5µm/OCT 10 µm)- Archived H&E/IF, USS, Block)
- Fiducial frame size/chemistry (6.5x6.5mm or 11x11mm, at 2x2µm bin)
- Sample QC
- Recommended slides for anti-detachment
- Experimental procedure (CytAssist)
- Library Chemistry (FFPE probe based /OCT cDNA synthesis)
- Data delivery (FATSQ and high res H&E/IF image) - needs SpaceRanger
- Loupe Browser
- Discovery technique
- No protein panel, but allows up to ≈ 3 mpIF on the same section prior to CytAssist run

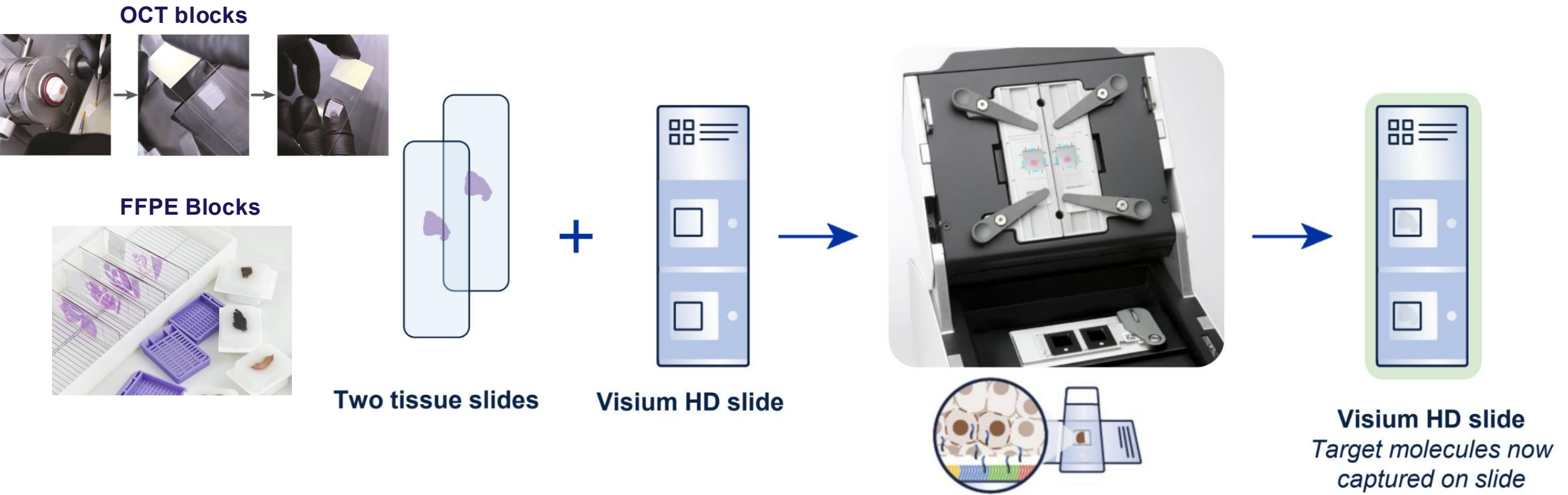
Classical Visium enabled 18,000 gene probe-based sequencing at a 55µm diameter (1-10 cell resolution/spot) with spots missing tissue every 55µm, covering a maximum of 5000 spots per 6.5x6.5 mm fiducial frame (both also available at 11x11 mm)



Visium HD slides contain two 6.5 x 6.5 mm Capture Areas with a continuous lawn of oligonucleotides arrayed in millions of 2x2µm barcoded squares without gaps, achieving single cell-scale spatial resolution. The data is output at 2 µm, as well as multiple bin sizes. The 8x8 µm bin is the recommended starting point for visualization and analysis

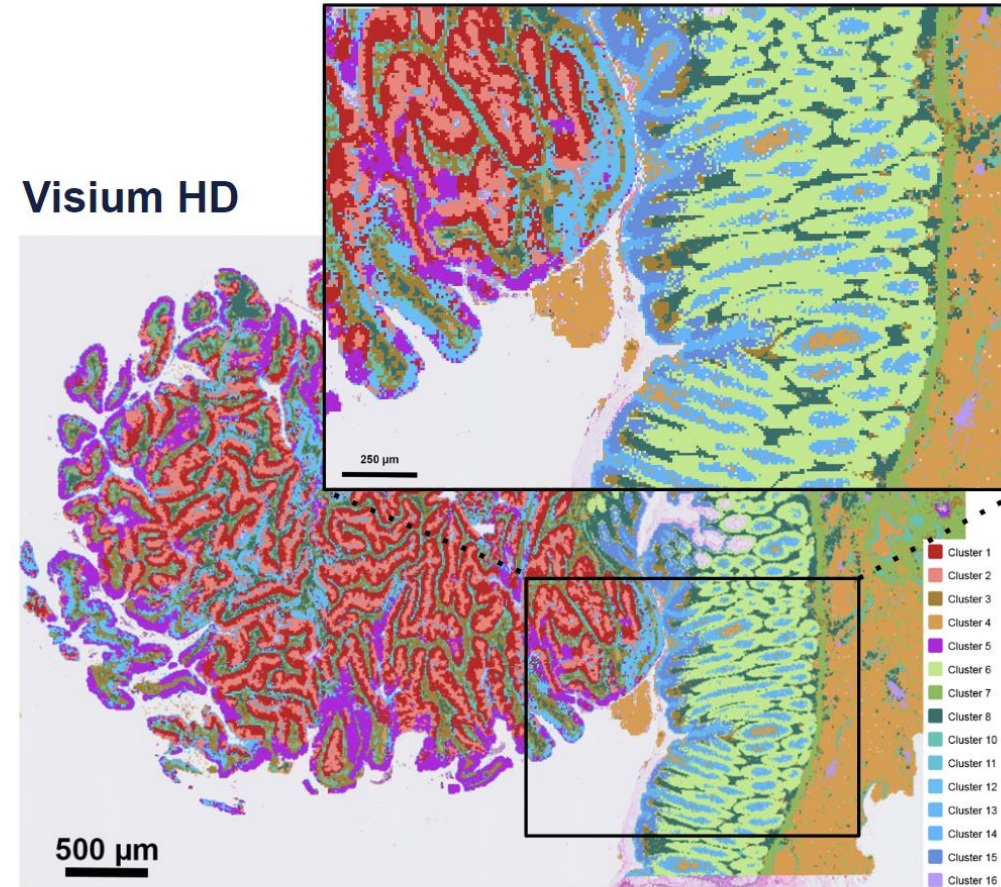
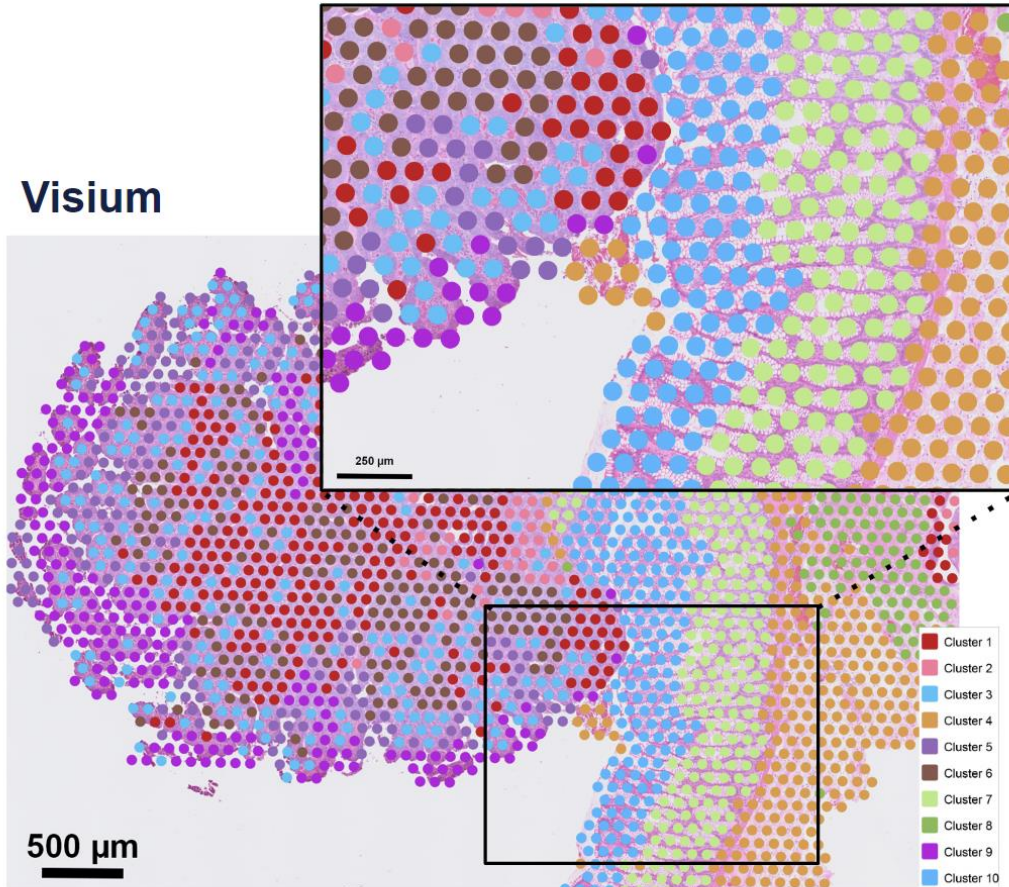


Researchers can start the workflow from **fresh frozen**, **fixed frozen**, or **FFPE** tissue blocks and freshly section onto glass slides, or from **pre-sectioned** tissue slides. After the H&E or IF staining steps, tissues are treated with whole transcriptome probe panels, enabling hybridization and ligation of probe pairs to their targets.



Tissue slides and Visium HD slides are loaded into the Visium CytAssist instrument, where they are brought into proximity with one another. Gene expression probes are released from the tissue, enabling capture by the spatially barcoded oligonucleotides present in a hydrogel on the Visium slide surface.

A comparison of Visium v2 data (left) and Visium HD data (right) in FFPE human colorectal cancer, demonstrating the enhanced discovery power of whole transcriptome spatial gene expression at single cell-scale resolution.



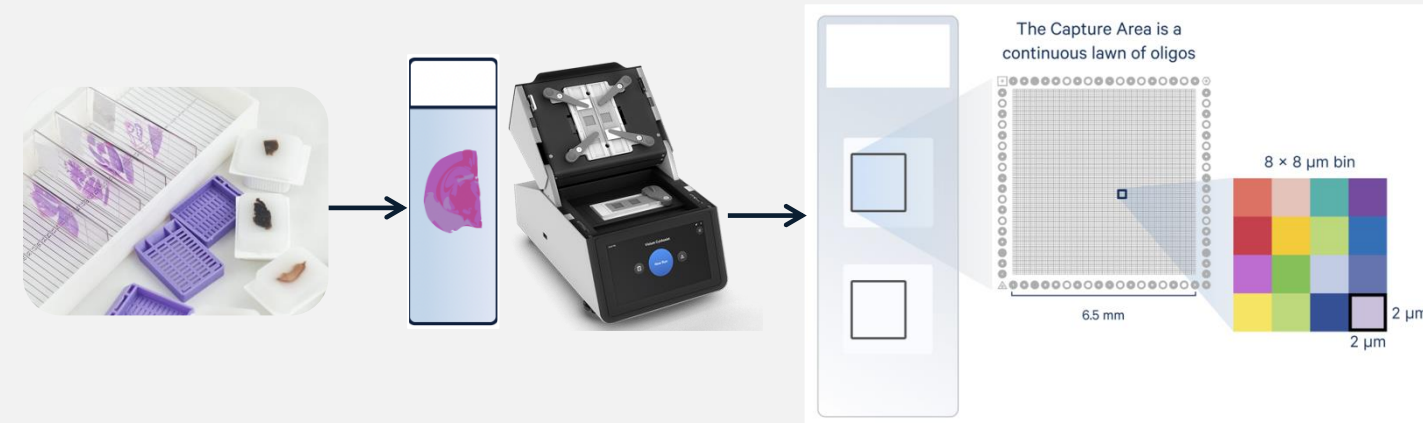
Visium Spatial Transcriptomics HD enables 18,000 whole transcriptome on 5 μ m sections of FFPE blocks or archived (un)stained slides with DV200>30% at a (2 μ m $\times$ 2 μ m) resolution

Input: 5 μ m FFPE/OCT tissue (6.5 $\times$ 6.5mm)

Output: high resolution H&E (brightfield) or IF (Fluorescent) image ($\approx$ 1GB) and FASTQ files

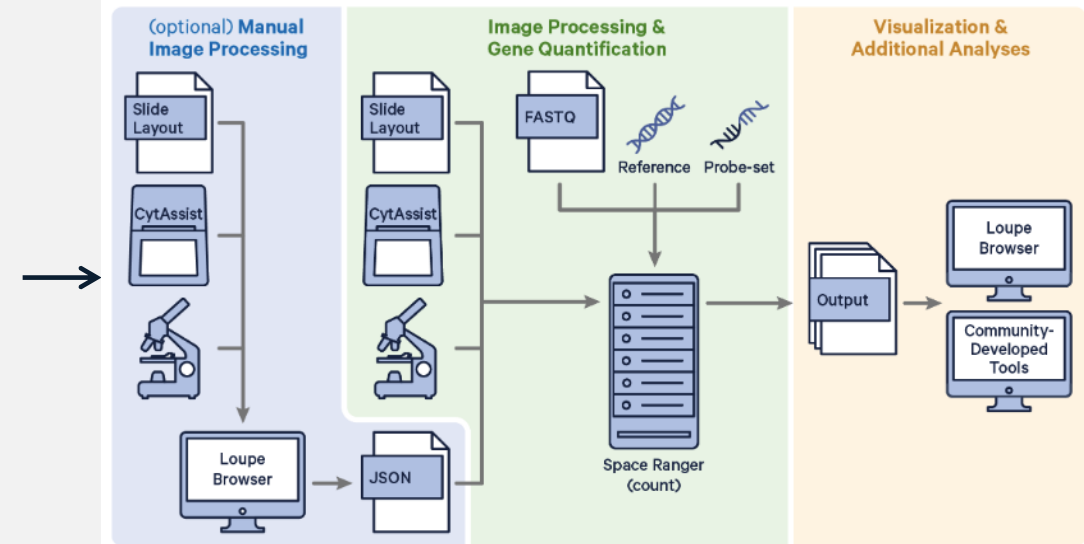
A

Visium HD experimental planning



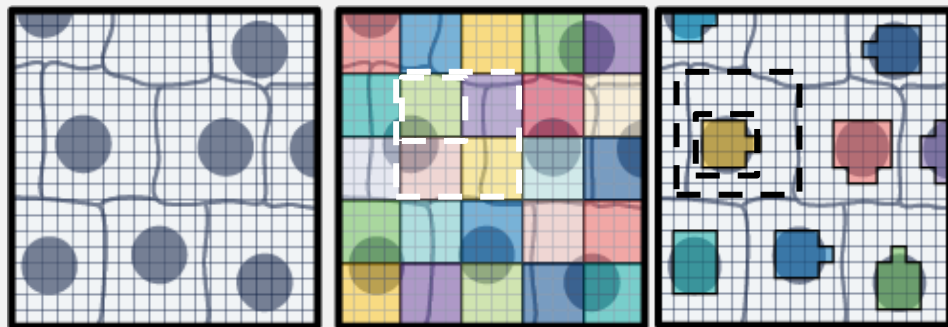
B

Space Ranger Alignment



C

Nuclei segmentation and custom binning

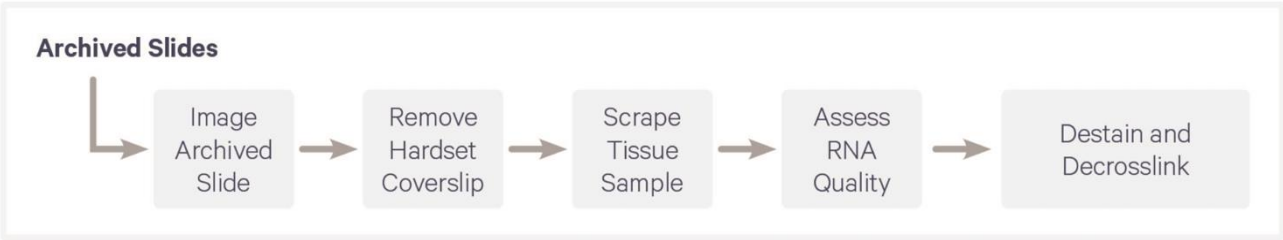


D

Third party tool analysis

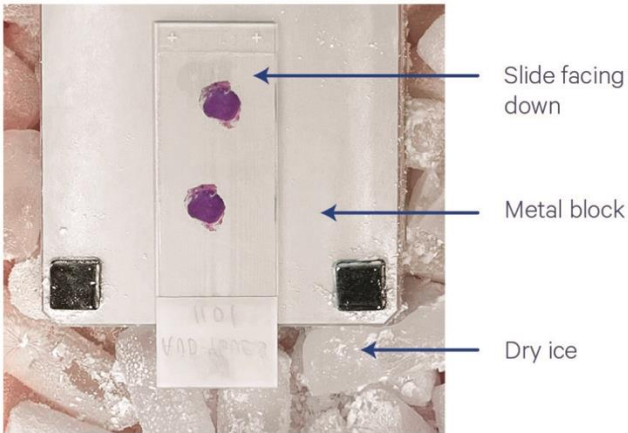


Using an archived H&E/IF stained slide for the Visium HD QC and CytAssist run



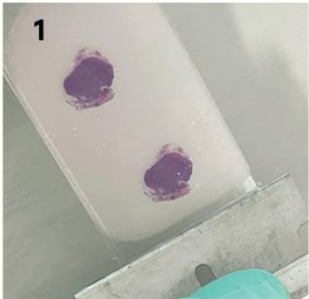
- Cool a metal block on dry ice for **5–10 min**.
- Gently immerse archived slide in Xylene Jar 1. Secure the jar cap to prevent xylene loss.
- Incubate for **5 min**.
- Remove excess xylene from archived slide with a lint-free laboratory wipe.
- Place on pre-cooled metal block with the coverslipped tissue sections facing down.
- Wait **1 min**.

Archived Slide on Metal Block

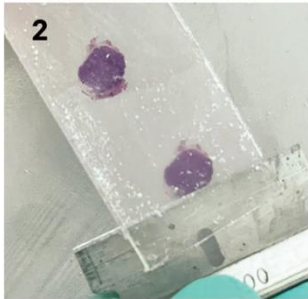


Removing Coverslip with Razor Blade

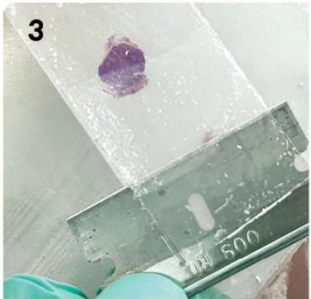
Align edge of razor blade to right most edge of coverslip



Slowly wedge razor blade between slide and coverslip until coverslip releases from slide

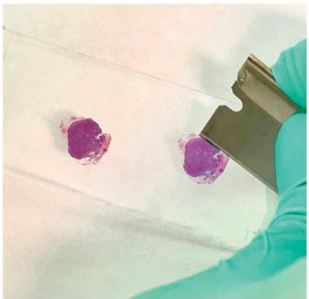


Holding razor blade towards coverslip, slowly work blade to the left to lift coverslip off slide

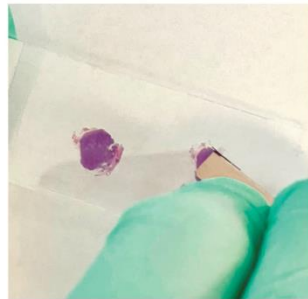


Scrape Tissue for RNA Quality Assessment

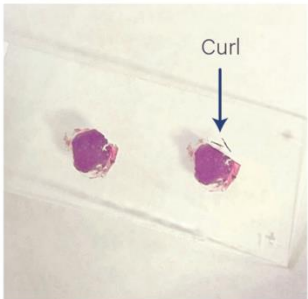
Scrape small portion of tissue at edge in one motion (tissue will curl on itself)



Cut off small portion of tissue from opposite side



Use small curl of tissue for RNA quality assessment



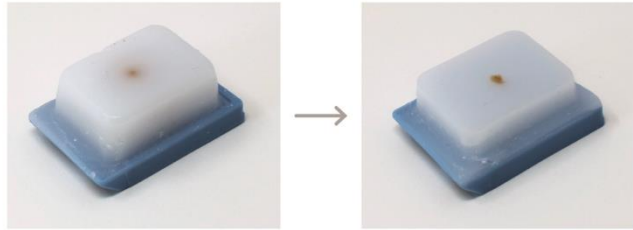
The remaining section on the archived slide may be used for the Visium HD workflow. If necessary, store the slide in a sealed slide mailer in a desiccator kept in the dark at 4°C for up to two hours to allow time for RNA evaluation.

Using an FFPE tissue block for the Visium HD QC and CytAssist run

FFPE Tissue Block



Exposing the Tissue Block



FFPE Tissue Block in Ice Bath

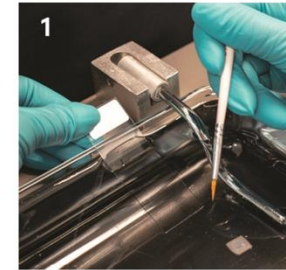


Remove Bubbles

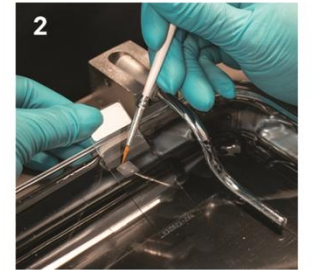


Section Placement

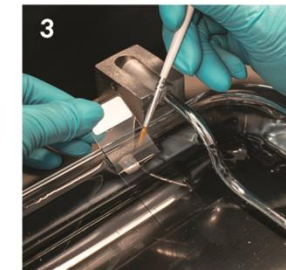
1. Section floating in the water bath



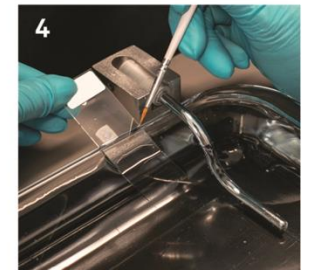
2. Blank slide inserted below surface of water



3. Align section edge within allowable area



4. Pull tissue slide out of the water



Workflow for Deparaffinization, H&E Staining & Decrosslinking

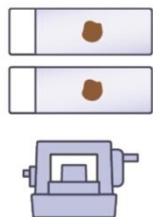
Sample Preparation

Deparaffinization

Staining & Imaging

Destaining & Decrosslinking

FFPE Tissue Sections
(human or mouse)



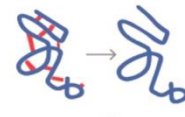
~3 h



Hematoxylin & Eosin
(H&E)



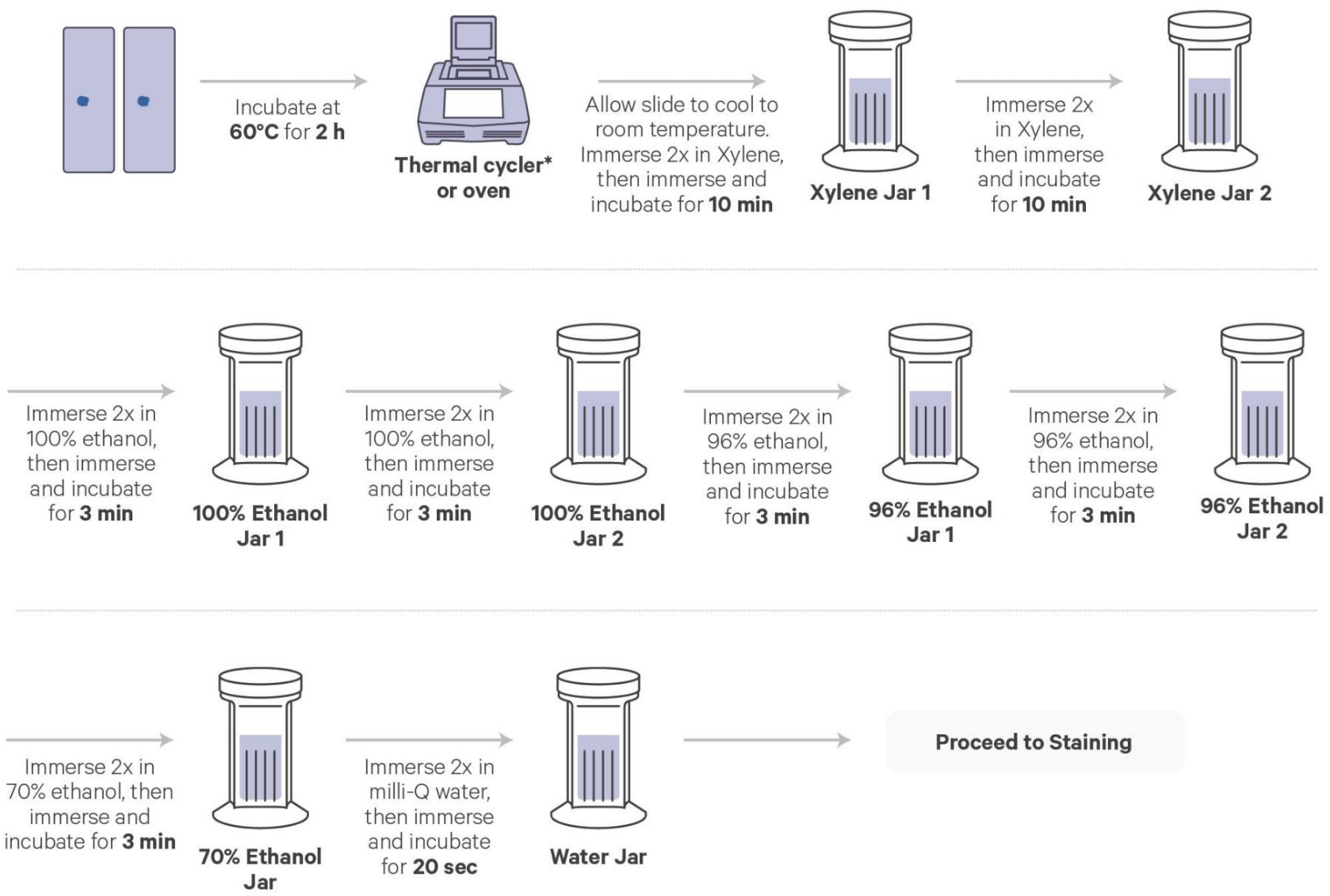
>1 h



~1 h

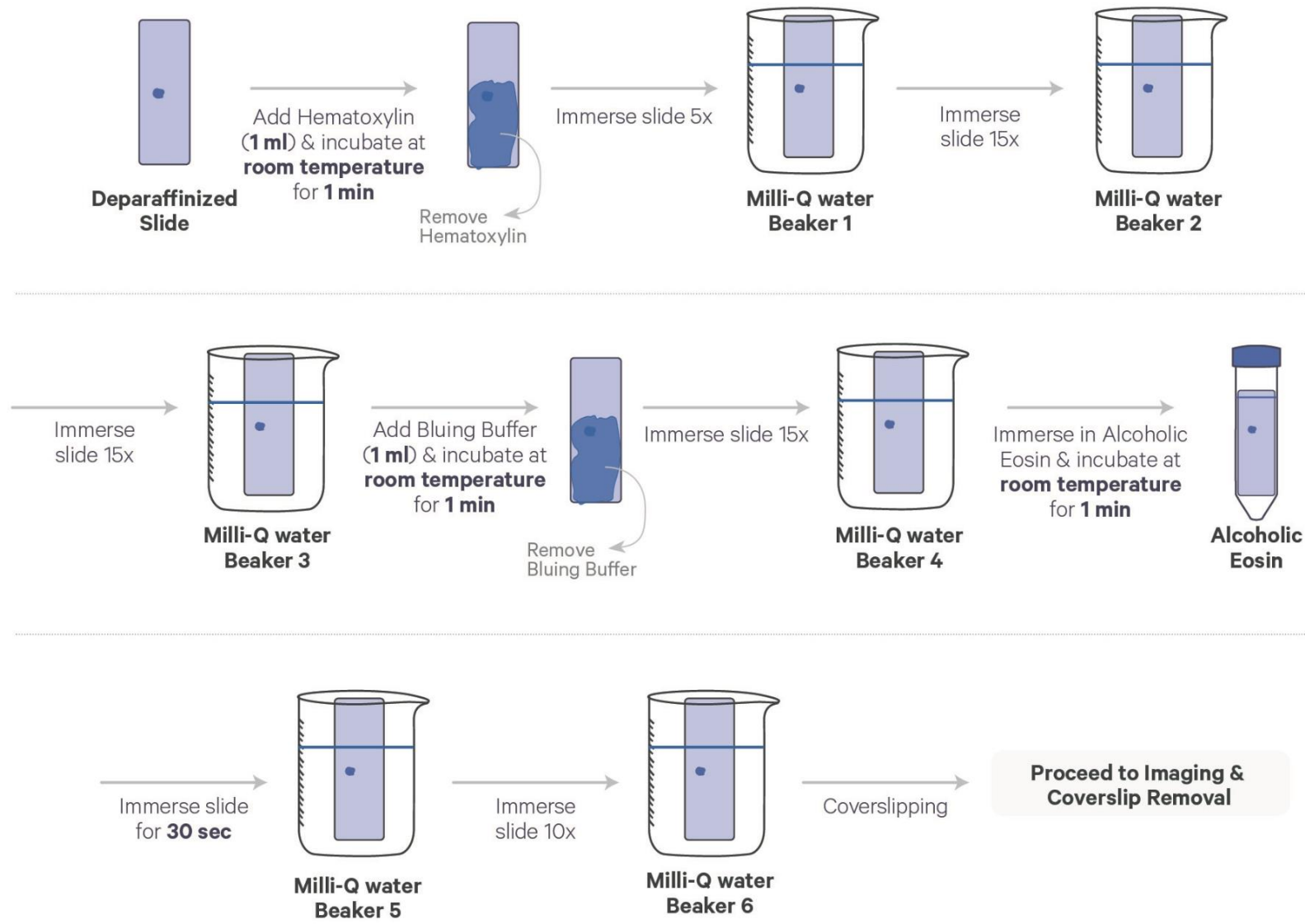
Proceed to
User Guide

Deparaffinization ~3 h



Slide in image is representative
*Keep the thermal cycler lid open during incubation

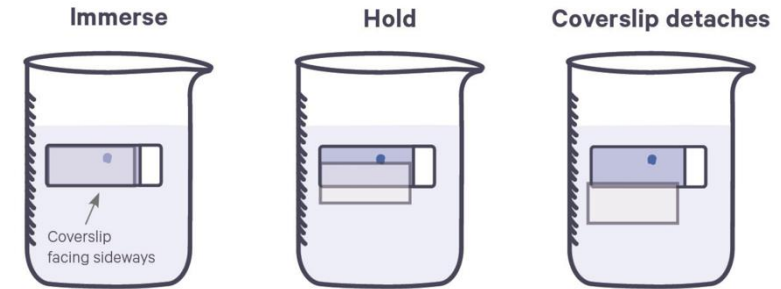
H&E Staining & Coverslipping ⌚ ~15 min



Cover slip removal:

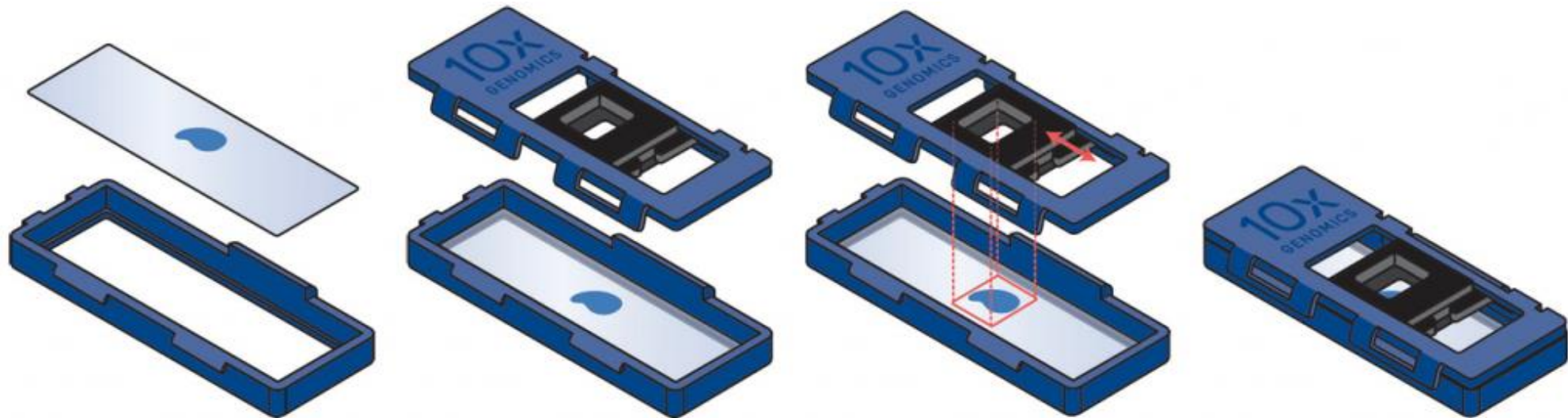
After high resolution microscopy image is acquired

Prior to de-staining and decrosslinking

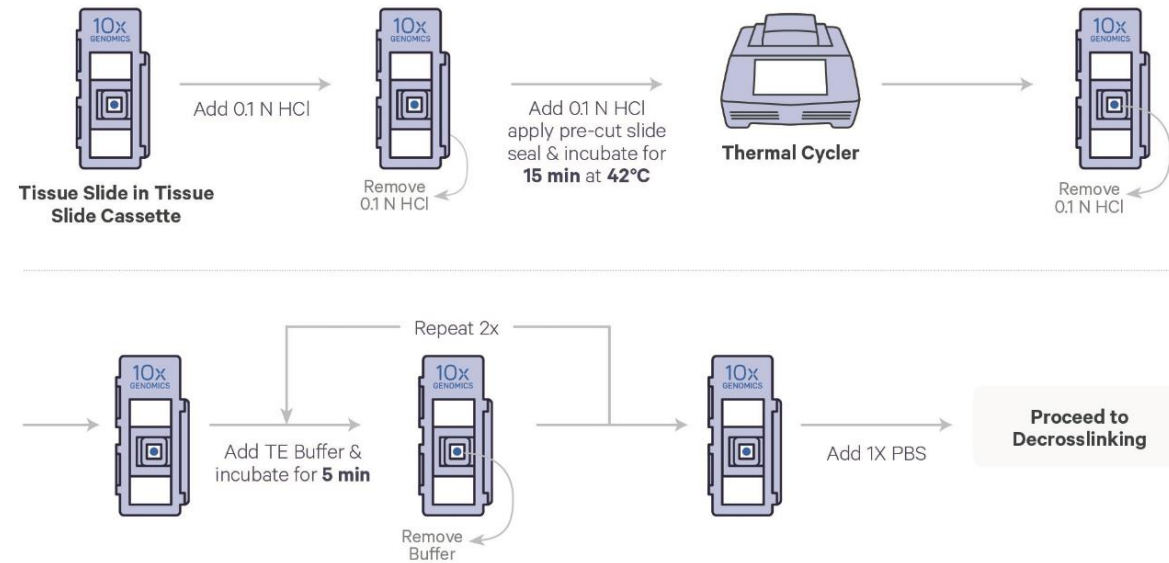


*Wipe excess liquid from the back of the slide after each immersion series

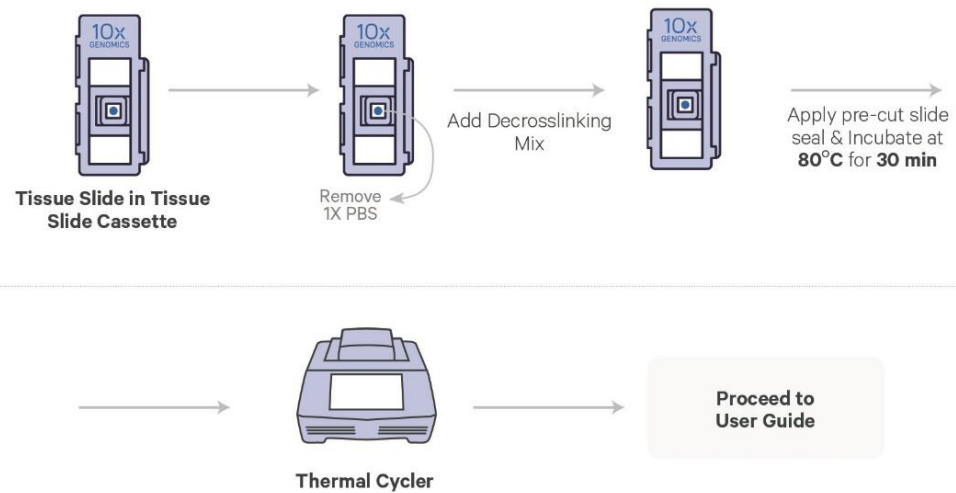
Fiducial alignment – Visium HD CytAssit Cassette



Destaining ~30 min

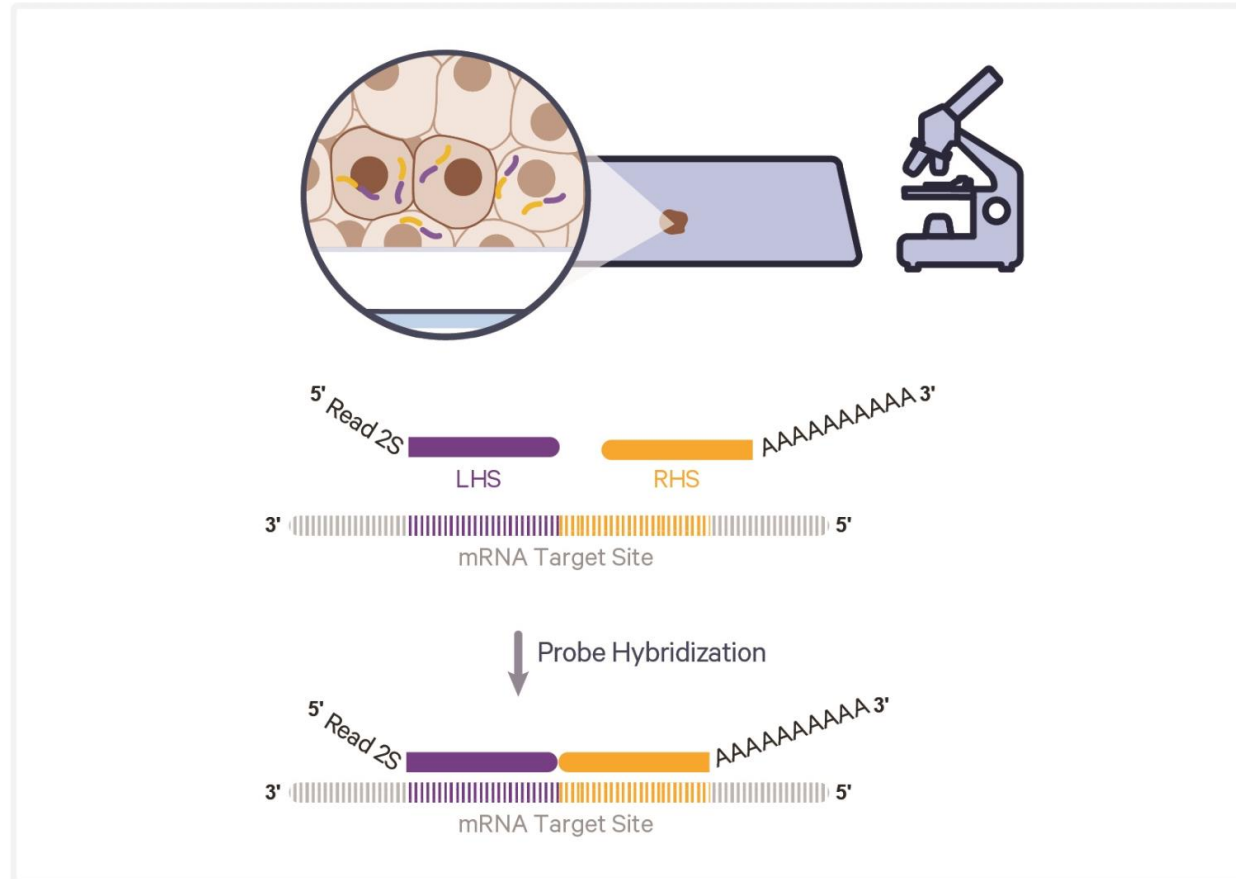


Decrosslinking ~30 min



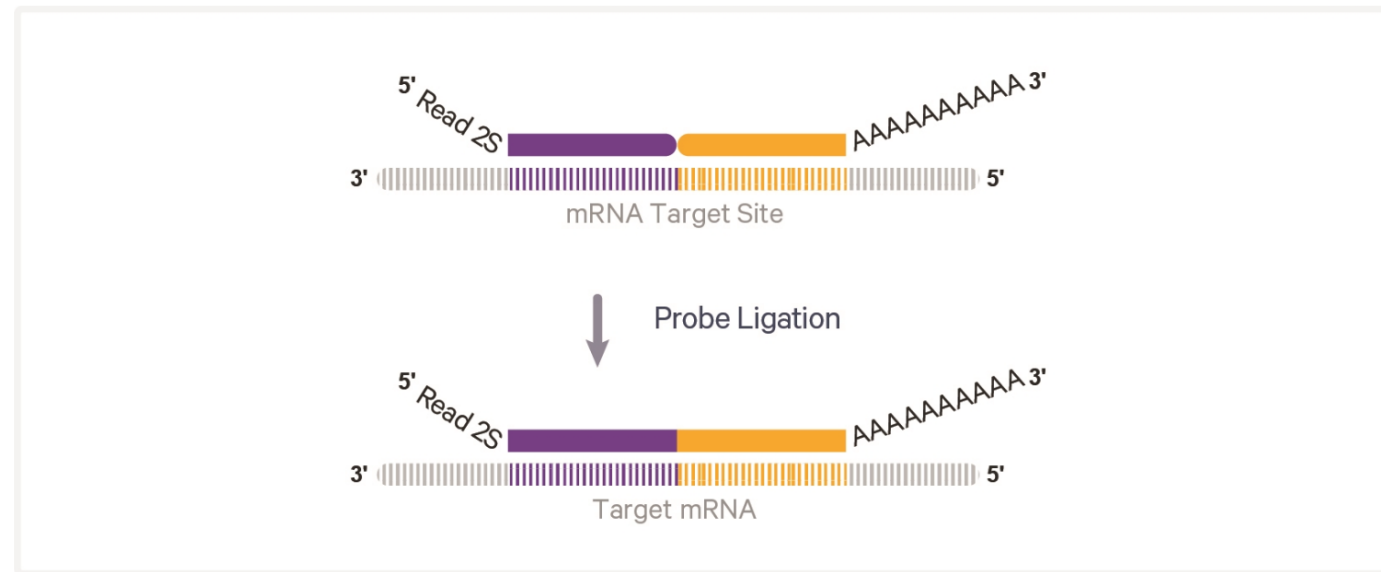
Step 1: Probe Hybridization

The human or mouse whole transcriptome probe panel, consisting of ~3 specific probes for each targeted gene, is added to the deparaffinized, destained, and decrosslinked tissues. Together, probe pairs hybridize to their complimentary target RNA.



Probe Ligation

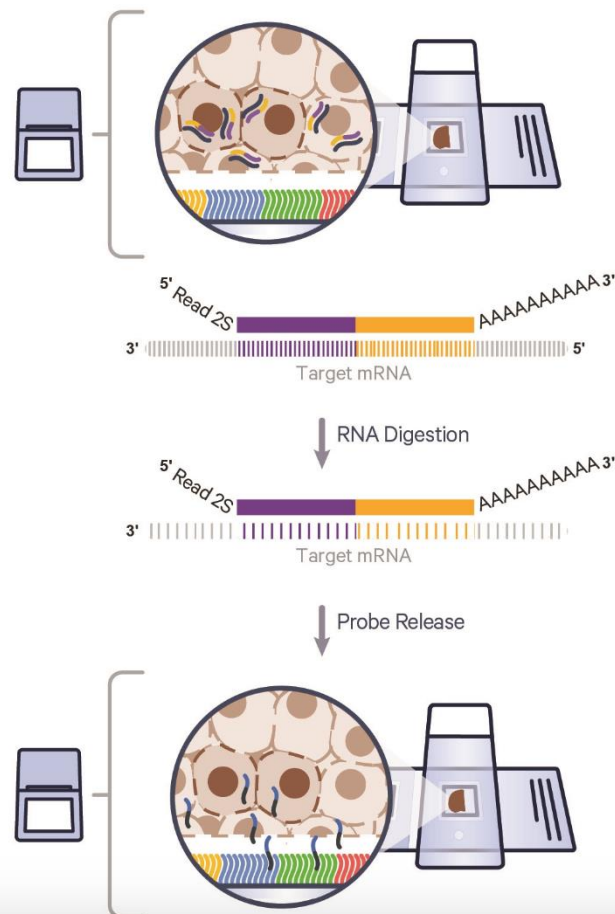
After hybridization, a ligase is added to bridge the junction between the probe pairs that have hybridized to RNA, forming a ligation product.



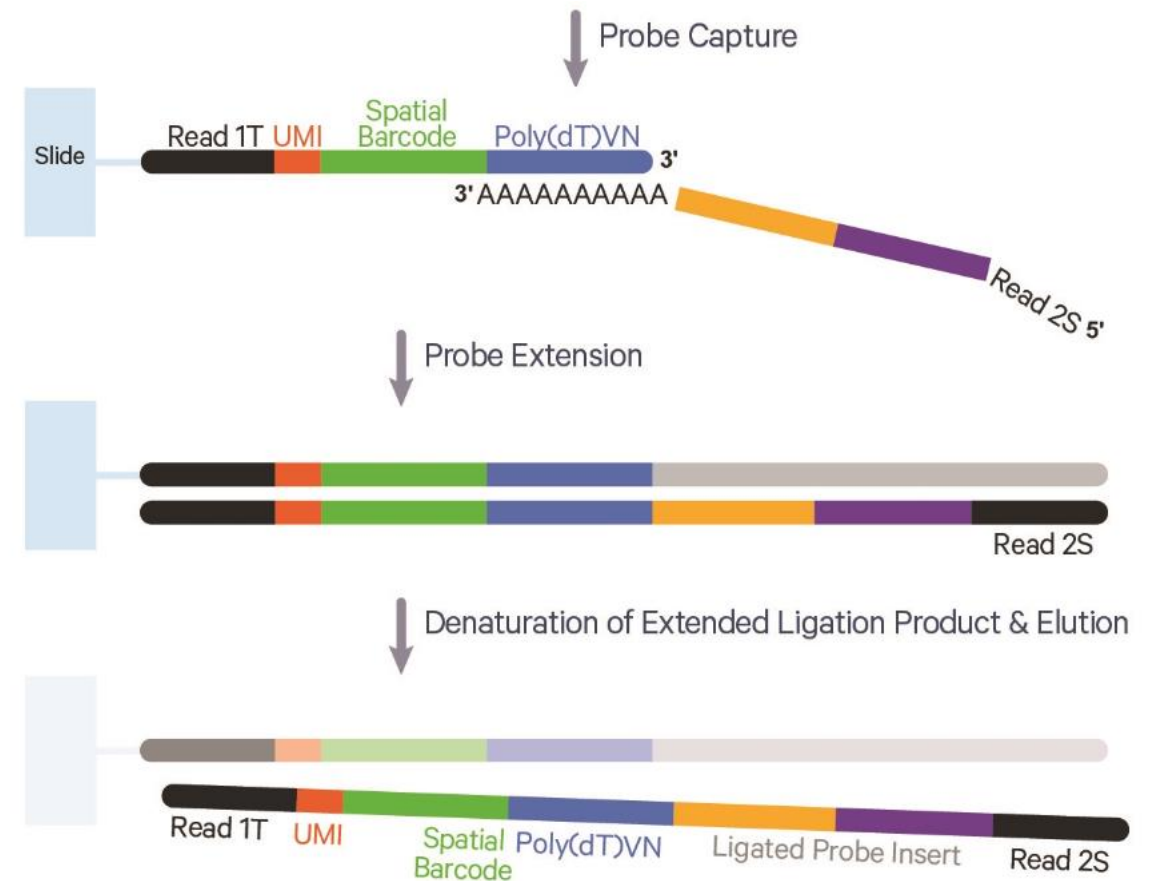
Probe Release & Extension

Probe release and capture occurs in the Visium CytAssist instrument. The single stranded ligation products are released from the tissue upon RNase treatment and captured on the Visium slide. Once ligation products are captured, the slides can be removed from the instrument.

Ligation products are extended by the addition of the Spatial Barcode, UMI, and partial Read 1 primer. This generates spatially-barcoded ligation products, which can then be carried forward for library preparation.

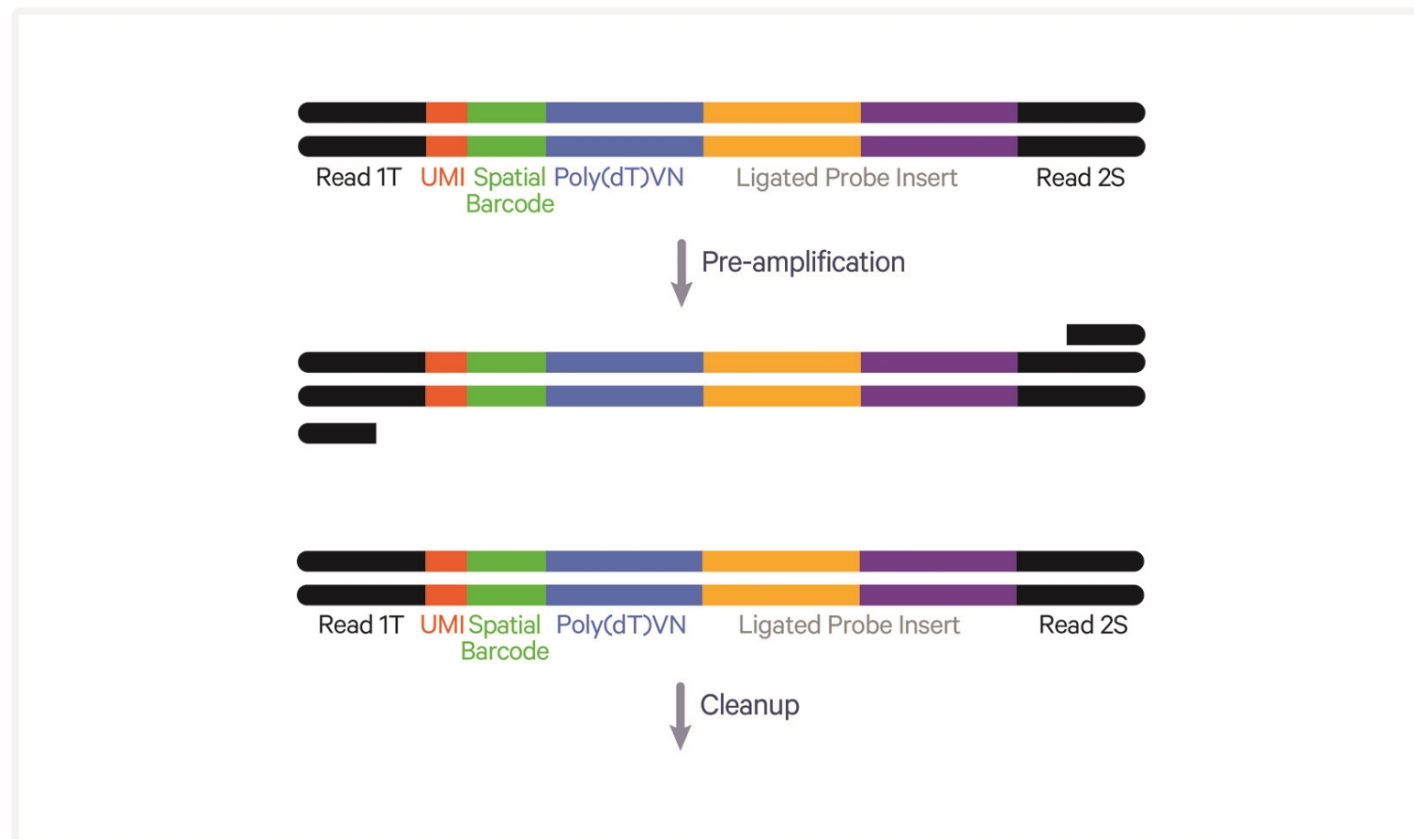


Visium HD Spatial Gene Expression Reagent Kits. CG000685 | Rev A



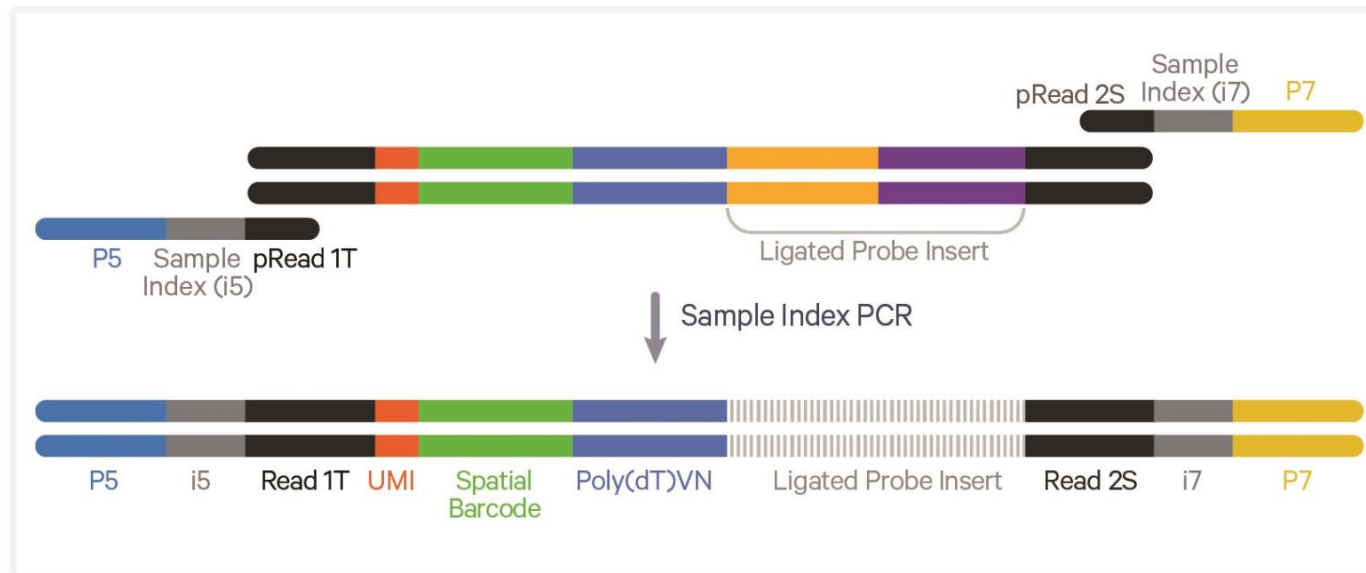
Pre-amplification and SPRIselect

To generate ample material for library construction, barcoded ligation products are amplified. This pre-amplification is followed by SPRIselect cleanup.



Step 6: Visium HD Spatial Gene Expression - Probe-based Library Construction

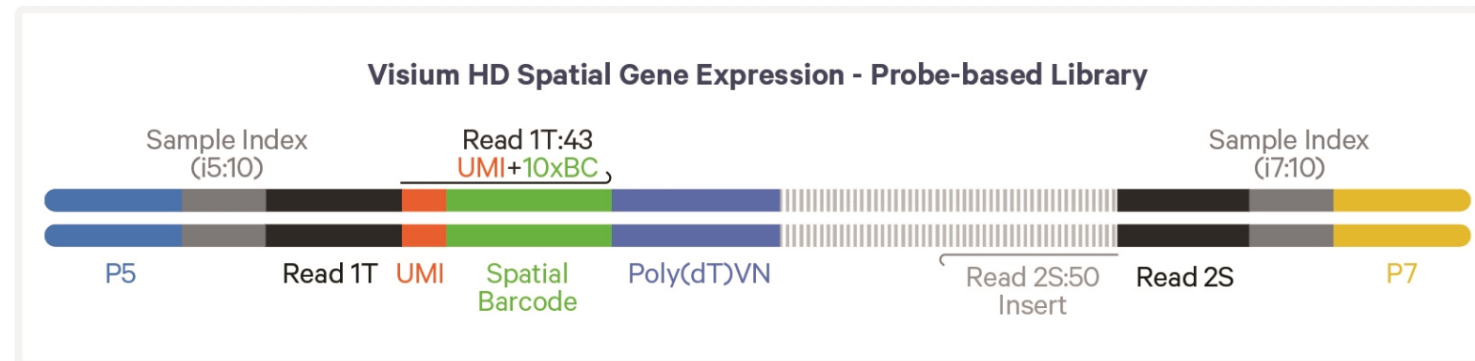
Pre-amplification material is collected for qPCR to determine sample index PCR cycle number for gene expression libraries. The amplified material then undergoes indexing via sample index PCR to generate final library molecules. The final libraries are cleaned up by SPRIselect, assessed on a Bioanalyzer or a similar instrument, quantified by qPCR, and then sequenced.



Sequencing

A Visium HD Spatial Gene Expression - Probe-based library comprises standard Illumina paired-end constructs, which begin and end with P5 and P7 adaptors. The 43 bp UMI and Spatial Barcode are encoded in TruSeq Read 1, while Small RNA Read 2 (Read 2S) is used to sequence the ligated probe product.

Illumina sequencer compatibility, sample indices, and library loading information are summarized in the Sequencing section.

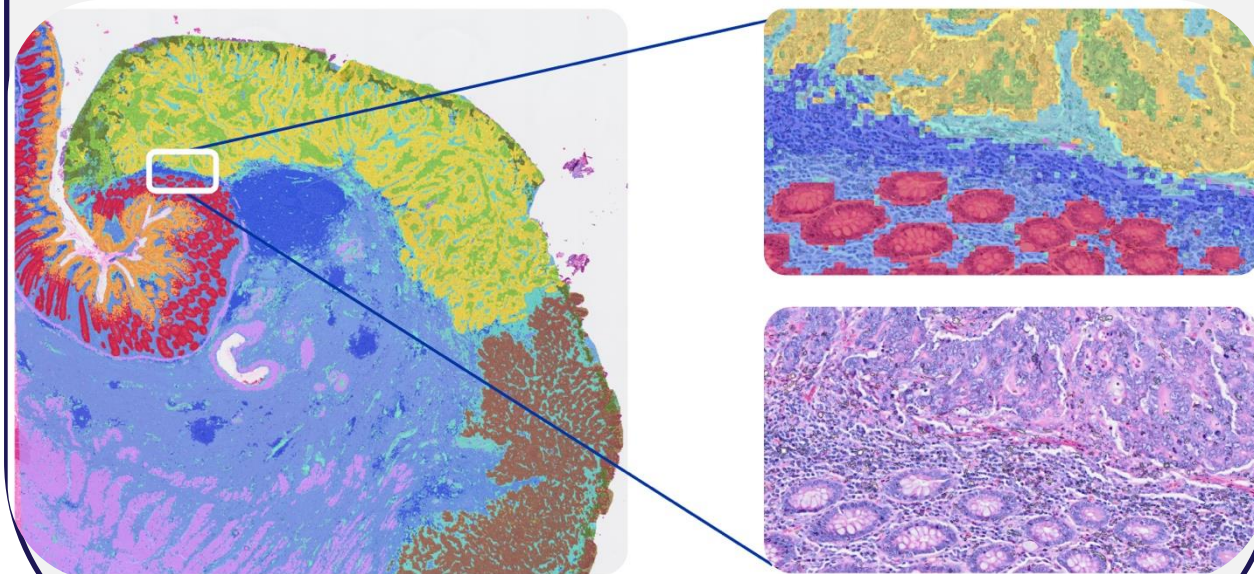
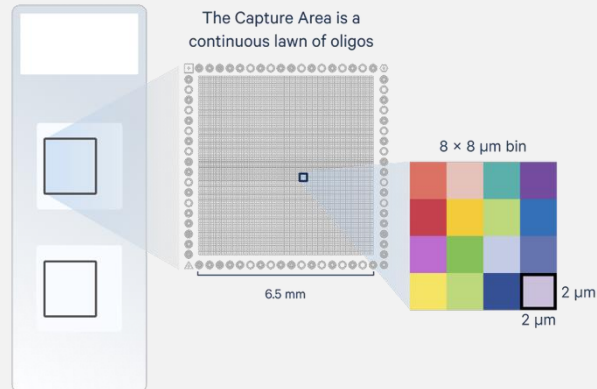


Xenium

- Material (freshly cut block on the Xenium Slide)
- Can add several samples per slide
- Non-destructive protocol, can perform H&E or IF or even Visium HD after the protocol is done
- Slide dimensions (22.5x10.5mm)
- Machine cycles, image-based sequencing
- Xenium Explorer, performs cell segmentation, and delivers pre-processed data
- Single cell/subcellular resolution (50nm)
- Add-on sets of protein panel (limited)
- Fully customizable panel (very expensive) and off the shelf panels for specific tissue/cancer types and a pan-cancer panel (500 more sensitive, 5000 available)
- Validation technique

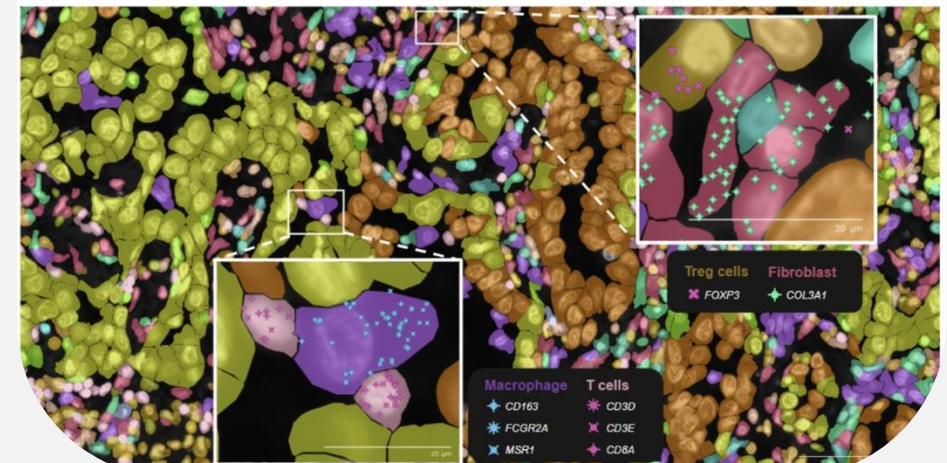
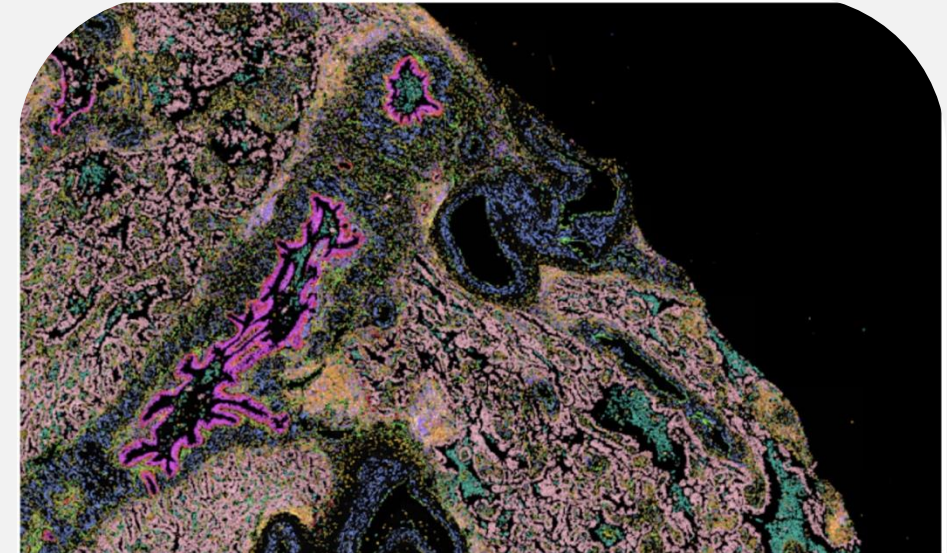
Visium HD

Visium HD on human colon cancer with unsupervised gene expression clustering and zoomed-in samples of clustered gene expression in $8 \times 8 \mu\text{m}$ bins



Xenium

Xenium Explorer analysis of human lung cancer with segmented cells and selected transcripts associated with immune cell types localized within individual cells.



Visium HD vs Xenium



Technical Specs

	Visium HD	Xenium Prime 5K
Methodology	Sequencing-based spatial transcriptomics	Imaging-based spatial transcriptomics
Data acquisition	NGS	High-resolution imaging
Analyzable area	6.5 x 6.5 mm per tissue section (2 per Visium slide)	22.5 x 10.5 mm per slide (2 slides per run)

Plexity

Number of RNA targets	Whole transcriptome (~20,000 transcripts)	Up to 5,000 genes
Customization options	Custom probe spike-in for custom and exogenous genes	Up to 100 custom genes added to pre-designed panels Advanced custom probes for exogenous genes and gene isoforms <i>Note: Standalone custom 480-plex panels are also available on Xenium</i>

Sensitivity

Transcript abundance (total transcripts per analyzed area)	~1.5-fold greater than Xenium Prime 5K*	Tissue dependent
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Visium HD vs Xenium

Resolution

	Visium HD	Xenium Prime 5K
Resolution	Single cell scale (data output at 2 x 2 µm and multiple bin sizes; 8 x 8 µm bin is a recommended starting point for analysis)	Subcellular (transcripts localized within single cells with < 50 nm (XY-localization) and < 100 nm (Z-localization) precision)
Cell segmentation	Requires third-party tools	Multimodal cell segmentation performed during instrument run

Multimomics

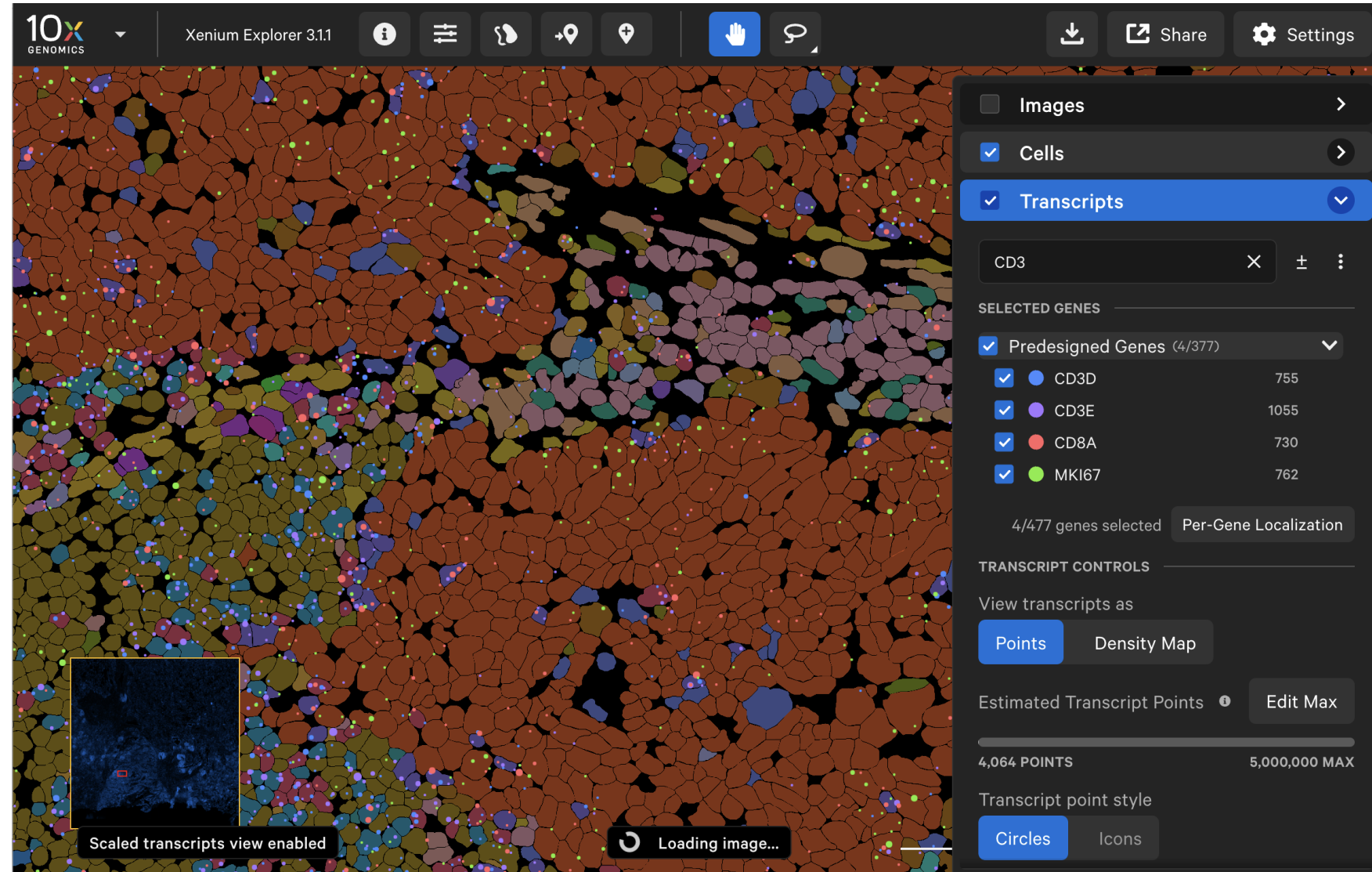
Additional assays on same section	Pre-run: Morphology (H&E) or IF	Post-run: Morphology (H&E), IF, and Visium/Visium HD
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Tissue Fixation

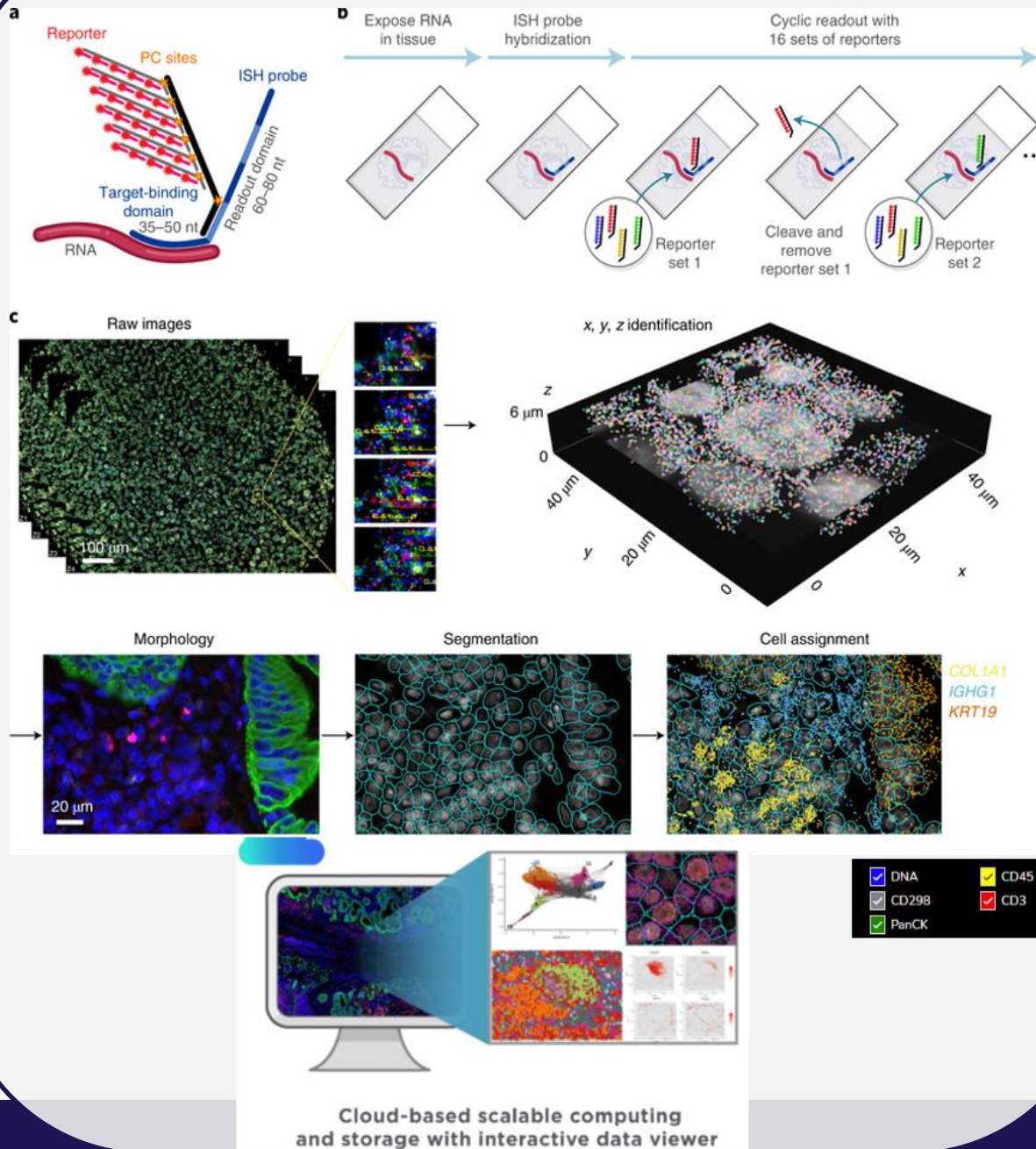
Sample compatibility	FF, fixed frozen, and FFPE	FF and FFPE
Organisms	Human and mouse, with non-human primate and rat possible depending on customer support guidance	Human and mouse <i>Note: Other species are available on standalone panels with up to 480 genes through Advanced Custom Design</i>
Tissue format	Pre-sectioned/archived slides Tissue block (sectioned onto standard glass slides)	Tissue block (must be sectioned onto Xenium slide)

Live example of Xenium Explorer

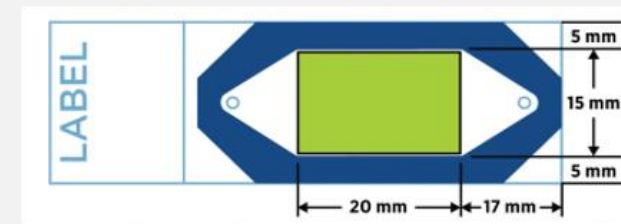
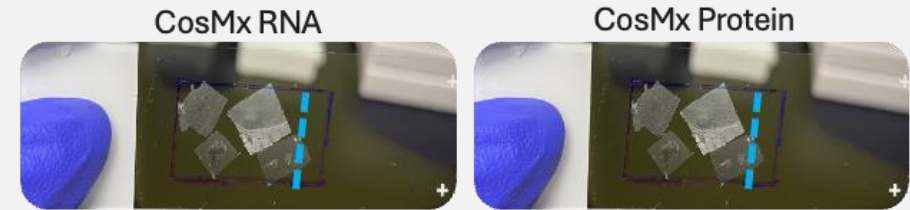
- > analysis
 - analysis_summary.html
 - analysis.zarr.zip
- > aux_outputs
 - cell_boundaries.csv.gz
 - cell_boundaries.parquet
- > cell_feature_matrix
 - cell_feature_matrix.h5
 - cell_feature_matrix.zarr.zip
- cells.csv.gz
- cells.parquet
- cells.zarr.zip
- experiment.xenium
- gene_panel.json
- metrics_summary.csv
- > morphology_focus
 - morphology.ome.tif
 - nucleus_boundaries.csv.gz
 - nucleus_boundaries.parquet
 - transcripts.parquet
 - transcripts.zarr.zip



- Material (freshly cut block on a few recommended slides)
- Can add several samples per slide
- Destructive protocol,
- Slide dimensions (20x15mm)
- Machine cycles, image-based sequencing
- AtoMx platform, performs cell segmentation, and delivers pre-processed data
- Single cell/subcellular resolution (50nm)
- No protein panel on the same slide, but available for adjacent tissue 64-plex vs 120 plex
- Can add a few customizable panel (very expressive) and off the shelf panels pan-cancer (1000 and 6000? And 19,000 plex)
- Validation technique



Storage at 4C after sectioning for RNA <2 weeks, and for protein <8 weeks
working area/slide = 15x20mm (≈4 samples)



Comparatively longer run time than other ISH techniques

CosMx™ SMI Turnaround Time Estimation

Analyte	Plex Format	Slide Format	Turnaround Time (Days)				
RNA	100	2-Slide	2	3	4	8	11
		4-Slide	2	4	8	-	-
	1000	2-Slide	3	4	7	13	-
		4-Slide	4	7	13	-	-
Protein	64	2-Slide	1	3	5	10	-
		4-Slide	2	5	10	-	-
Tissue Area Coverage			16 mm ²	50 mm ²	100 mm ²	200 mm ²	300 mm ²
Fields of View			62	193	386	772	1158

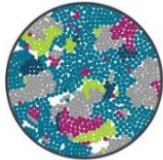
Turnaround estimates are designed as experimental guidance and may change slightly from run to run.

FOR RESEARCH USE ONLY. Not for clinical use.

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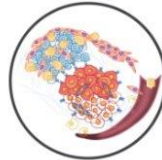


Key Applications



Cell Atlasing/Cell Typing

Discover and map cell types using expression profiles of known RNA and protein targets



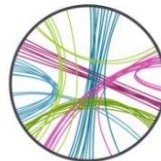
Disease State

Visualize and quantify molecular (RNA / protein) and cellular organizational changes in a tissue



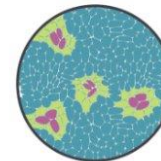
Ligand-Receptor Interaction

Analyze expression and interactions of up to 100 classic ligand-receptor pairs between interacting cells



Biomarker Discovery

Reveal differential gene expression and pathways in the same cell types depending on their spatial location



Tissue Microenvironment

Understand cellular neighborhoods by examining individual cells and their interacting neighboring cells

CosMx AtoMx Platform

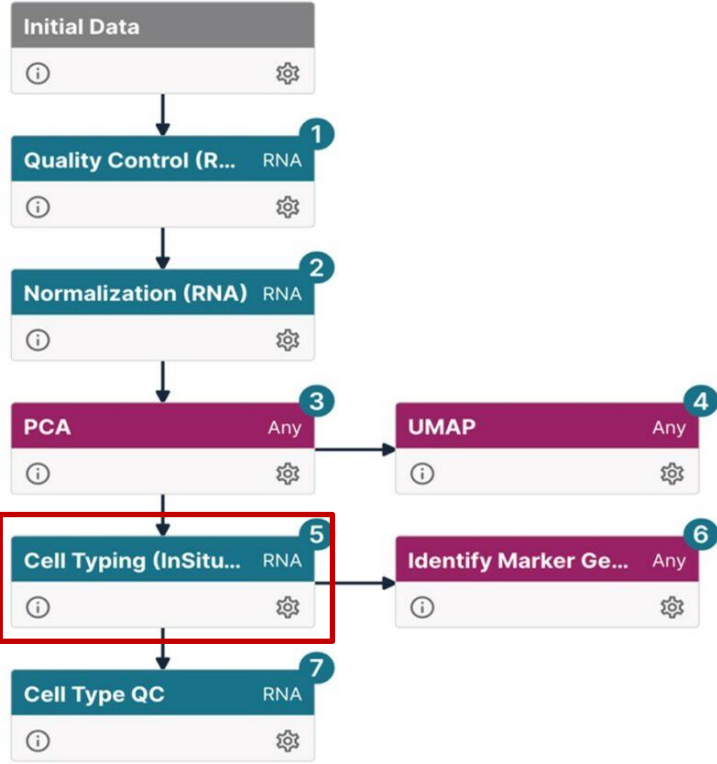
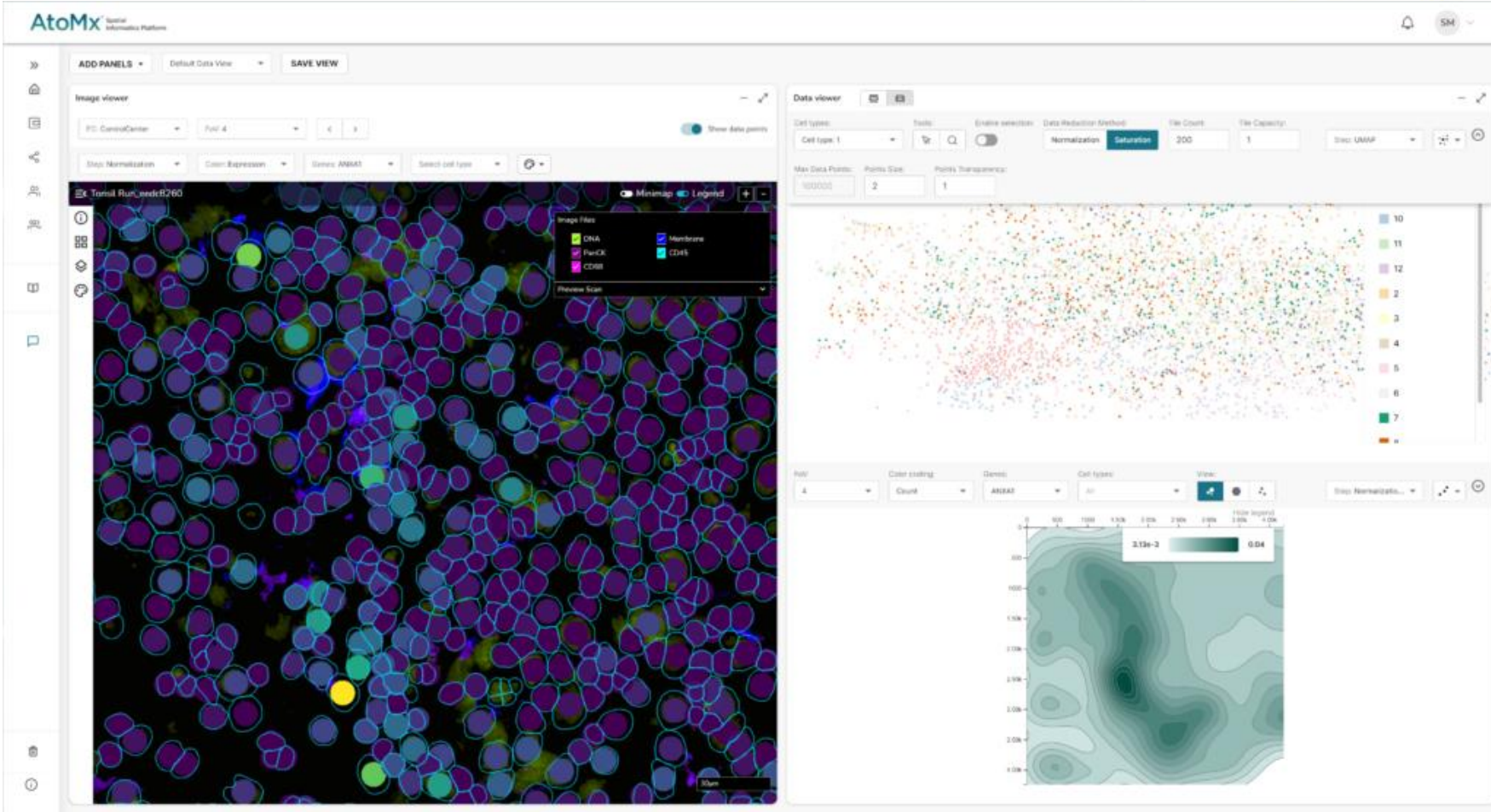


Figure 1: the standard AtoMx cell typing pipeline for CosMx RNA assays.



Can add publicly available gene/cell type matrix from scRNA data to transfer cell type labels